$$x \in \mathbb{R}^{H \times W \times 3}, \quad z = \mathcal{E}(x) \in \mathbb{R}^{h \times w \times c}$$

$$h = H/f, \quad w = W/f, \quad \hat{x} = \mathcal{D}(z)$$

$$256 \times 256 \times 3 \xrightarrow{f=8} 32 \times 32 \times 4$$